

Homework — 1.2.4 Types of Programming Language

Name: _____ Date: _ **Mark:** ____ / 20

OCR H446 — set after the 1.2.4 lesson. Show your working.

Part A — This week's topic (~14 marks)

1. Explain what is meant by a **programming paradigm**, and explain why no single paradigm is best for every problem. **[2]** (AO1)

2. State **three** features that characterise the **procedural** paradigm. **[3]** (AO1)

3. The following Little Man Computer (LMC) program reads two numbers and outputs a result.

```
    INP
    STA first
    INP
    SUB first
    BRP positive
    OUT
    HLT
positive OUT
    HLT
first DAT
```

(a) Describe in words what value this program outputs. **[2]** (AO2)

(b) State what the **BRP** instruction does and why it is used here. **[2]** (AO1)

4. Addressing-mode calculation. A processor executes **LDA 4**. The relevant memory contents are:

Address 4 holds the value **7** · Address 7 holds the value **2** · The index register holds **3**.

State the value loaded into the Accumulator for each mode below. **[3]** (AO2)

(a) Immediate: __ (b) **Direct**: _ (c) **Indirect**: ____

(d) State the *effective address* used in **indexed** addressing: _____

5. Explain the OOP concept of **inheritance** and give **one** benefit it provides to a programmer. **[2]** (AO1)

Part B — Retrieval: keep it fresh (~6 marks)

6. (from 1.2.3) State **two** advantages and **one** disadvantage of the **Waterfall** methodology. **[3]** (AO1)

7. (from 1.2.2) Explain the difference between **static** and **dynamic** linking. [2] (AO1)

8. (from 1.1.1) State the purpose of the **Accumulator (ACC)** register. [1] (AO1)